Individual Evidence Pack: Milestone 3

Student Name: WU Wanpeng
Role: Member A - Architecture Lead & Spec Owner
Project: PolyWatch - Polymarket Integrity Monitor
Date: Mar 29, 2026

---

1) What I contributed since Milestone 2

- Completed final architecture-to-implementation traceability review, mapping threat items (T-02/T-03/T-05) to existing modules and executable specs.
- Consolidated final submission evidence into a single reviewer-friendly structure (artifact links + reproducible validation output).
- Performed regression validation on peer modules `ZScoreDetector` and `WhaleAlert` to verify behavior stability before final handoff.
- Recorded peer challenge and my response on anomaly threshold calibration (false-positive control for production).

---

2) Verifiable evidence (at least 2 items)

Evidence A — Design artifact: Threat model update used in final phase

- File: `milestone_2/threat_model_v2.md`
- Verifiable content:
  - T-05 (Benford statistical fabrication) was added
  - T-02 impact was upgraded to HIGH
  - T-03 metric was refined with block-timing correlation

Evidence B — Design/Spec artifact: Executable Benford specification

- File: `milestone_2/specs/benford_law.feature`
- Verifiable content:
  - PASS/FAIL/INSUFFICIENT-DATA scenarios
  - Configurable chi-square threshold scenario
  - Boundary-condition `Scenario Outline` with example table

Evidence C — Validation output summary (Mar 29, 2026)

- Scope: Peer modules in `core_analysis/zscore_detector.py` and `core_analysis/whale_alert.py`
- Reproducible output:

```
VALIDATION_A_ANOMALY_COUNT= 4
VALIDATION_A_TOP_INDEXES= [74, 106, 113, 120]
VALIDATION_A_MAX_Z= 3.939577
VALIDATION_B_WHALE_COUNT= 3
VALIDATION_B_WHALE_IDS= [2, 5, 8]
VALIDATION_B_WHALE_SIZES= [300, 410, 500]
```

Evidence D — Peer feedback/challenge and response

- Challenge from peer review: `ZScoreDetector` at threshold = 2.5 produced extra statistical flags under Gaussian noise windows.
- My response: Keep 2.5 for high-sensitivity MVP detection, but document production tuning option to 3.0 to lower false positives; track as post-milestone calibration item.
- Supporting context already documented in: `milestone_2/evidence_pack_milestone2_memberA.md`

---

3) Validation I performed (at least 1 item)

Validation Item — Regression verification of peer algorithm modules

What I tested/verified:

- Verified `ZScoreDetector(window=20, threshold=2.5)` still catches injected spike event.
- Verified `WhaleAlert(size_threshold=200)` still returns all and only expected whale trades.

Result:

- `ZScoreDetector`: 4 anomalies detected, including injected spike index `120` with max z-score `3.939577`.
- `WhaleAlert`: 3 whale trades detected (IDs `2, 5, 8`; sizes `300, 410, 500`).
- Conclusion: No regression observed in current code behavior for these test fixtures.

---

4) AI usage transparency (short)

Rejected AI output example + why:

- Rejected suggestion: “Lower Benford minimum sample size from 100 to 30 to increase alert coverage.”
- Why rejected: For chi-square goodness-of-fit, too-small sample sizes make digit-frequency inference unstable and increase false alerts; this would reduce evidentiary reliability in forensic reporting.